TransTech Systems, Inc.



PQI 380go
Pavement Quality Indicator

Operator's Handbook

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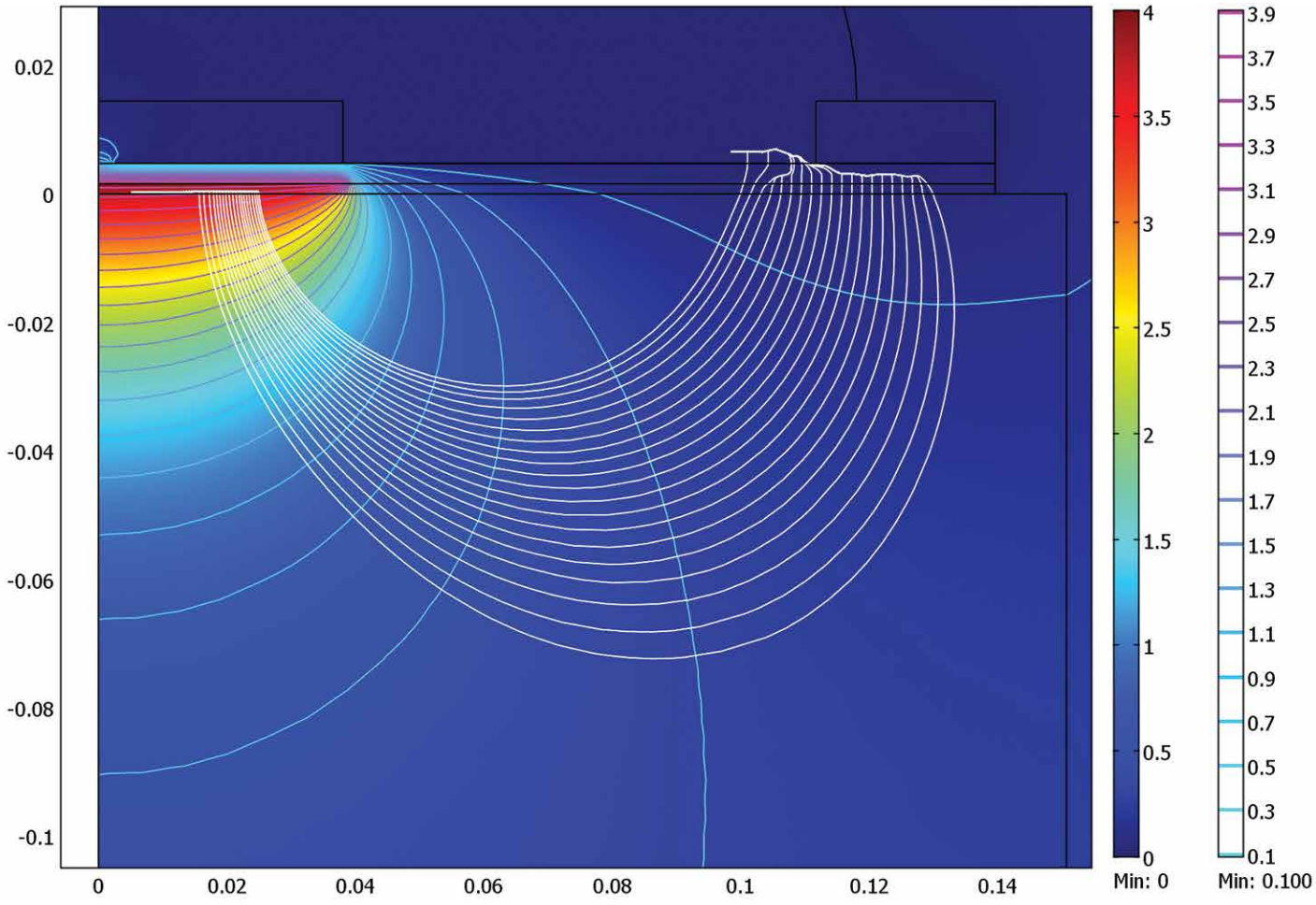
Introduction

TransTech's Pavement Quality Indicator model PQI 380go utilizes state of the art technology to achieve accurate asphalt pavement density readings. It's primary features are:

- Non-Nuclear Technology - No special license or radioactive materials required
- Lightweight and easy to use
- 7.5 hours of portable operation
- Measures density in common units (lb/ft^3 or kg/m^3)

Measurement Technology

The PQI 380go uses impedance spectroscopy to measure the electrical response of asphalt from which the density is calculated. As shown in the Comsol simulation below, the electrical field transmits through the material from the sensor plate of the PQI 380go. The impedance is then measured and used in the calculation of the density for that specific aggregate.



Application Summary

The PQI 380go is intended primarily for use on newly-laid asphalt pavements with lift thickness ranging from 0.75 inches to 6 inches. **The calibration of the PQI 380go to a density standard (core) is very important for the reliability and consistency of the density measurements for each material.**

Safety

Every effort has been made to make the PQI 380go convenient to use and inherently safe.

The PQI 380go uses non-nuclear, low-voltage direct current to obtain measurements, therefore, there are no badges, licensing, storage or transport concerns. Like any instrument, however, the user should exercise care and common sense in its use to prevent mishaps.

Warning

Do not use the unit on or near electrical wiring.
A potential shock hazard exists if contact is made with the exposed wiring.

Warning

Use care in handling the unit. Personal injury can occur through improper handling.
Take proper care to avoid accidentally dropping the unit.

Warning

Unauthorized disassembly of the unit will void the warranty.

Warning

Shipping the gauge with the batteries inside is not recommended.

Caution

Charging the gauge overnight or unattended is not recommended

Caution

Turn the unit off when not in use and during transport.

Caution

Be sure not to switch standardization plates with other gauges. Check the Serial Numbers located on the gauge and on the plate to be sure they match.

Disclaimer: TransTech Systems reserves the right to change or modify product design, construction, specifications, or materials without prior notice and without incurring any obligation to make such changes and modifications on TransTech Systems products previously or subsequently sold.

Controls & Components - Contents

The PQI 380go is packaged and shipped with the following components.
Contact TransTech Systems customer service if any of the parts are missing.

- PQI 380go Unit
- Standardization Plate
- Storage/Shipping Case
- Operators Quickstart Guide
- PQI 380go Handle
- 120/220V AC to 12V DC Battery Charger
- 12V DC Car Charger
- Battery Pack



Case



PQI 380go



Charger



DC Cord



AC Cord



Handle



QuickStart Guide

Gauge Features



Handle Extension
Connection for handle extension

On / Off Button
Powers gauge ON / OFF

Touch Screen
Used to input and interact with the gauge

Handle

Faceplate
Protective bezel for touchscreen



USB Port
Used for diagnostics and software updates

Charger Port
Allows connection to the ac or dc charger

Gauge Shell
Body of the gauge

Battery Door
battery cover plate

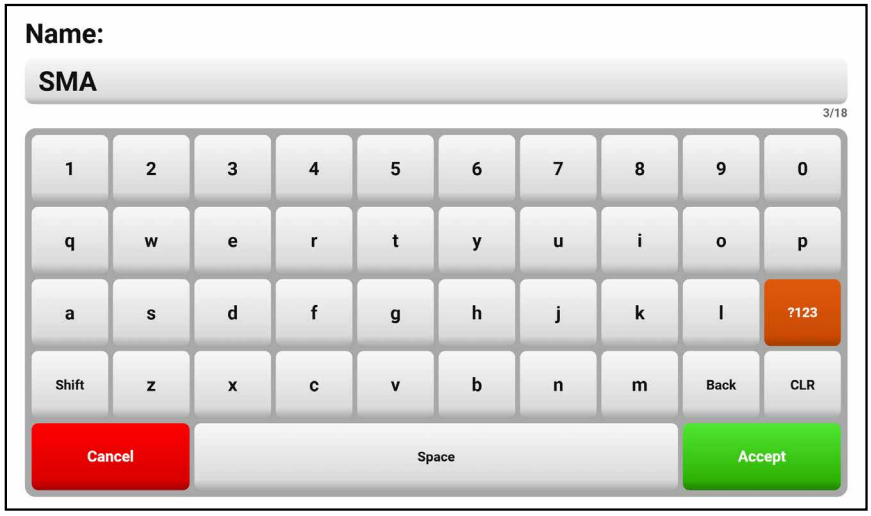
Shroud with 10” Sensor Plate
Location where sensor plate is located

External / Internal Controls



External controls on the PQI 380go consist of an ON/OFF switch and a 7” high-brightness capacitive touchscreen for navigating through the user interface and entering alpha/numeric data.

Be sure not to drag your finger from one button to the next. A firm yet light touch is all that is needed to navigate from screen to screen.



Unique yet consistent icons are located on most screens for easy one touch navigation. When data input is required of the operator the screen will display the above keyboard. Toggle from upper to lower case letters using the **Shift** button. Once editing is completed, tap **Accept** to save and return to the previous screen.

Status Bar

The **Status Bar** is located at the top right of the screen (excluding the keyboard screens) and will continuously update the remaining voltage of the battery while displaying the date, time as well as the current mix on measurement screens.



Power Save - Auto Shut Down

If the PQI 380go is left on for an hour without any buttons being pressed, it will shut itself down, to conserve battery run time.

Setting up the PQI 380go

Prior to using the PQI 380go for the first time the gauge will need to be configured to take measurements correctly.

The following steps must be completed before operating the PQI 380go:

1. Installing and charging the battery
2. Starting the software
3. Set the local time / set date
4. Select units of measurement
5. Define the asphalt being tested (Mix)
6. Select measurement mode (Single or Continuous)

Installing Battery



You will find a plate on the left side of the PQI 380go with 4 spring loaded bolts.

1. Turn each bolt approximately 2 full turns until the spring releases
2. Remove the plate from the gauge
3. Next you will find a wire with a connector, disconnect the battery from the gauge
4. Then pull the nylon strap to slide the battery out of the compartment.
   (Do not pull the battery by the connector wire, doing so could damage the battery)

Charging Battery

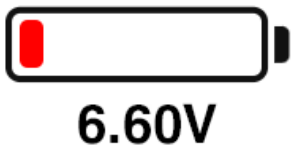
1. Turn the PQI 380go unit OFF
2. Connect the charger to the charger connector located on the back of the PQI 380go
3. Plug the charger into a standard AC outlet, verify red LED indicator is on.
4. The red indicator lamp will turn green to indicate that the battery is charged (approx. 4hrs)
5. Unplug charger from the power source before disconnecting the charger from the gauge.

Battery Care Tips

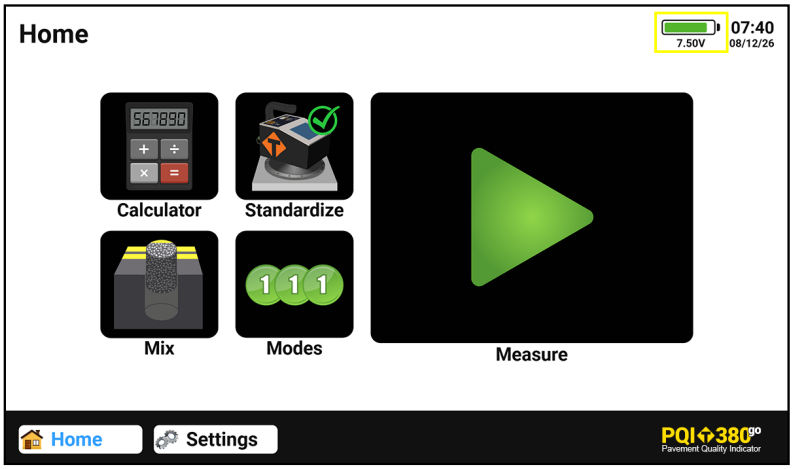
- Whenever uncertain about the battery charge level or condition, recharge it
- The battery will self-discharge and should **NOT** be left uncharged for more than 30 days
- An occasional complete discharge followed by a full recharge is recommended
- **NEVER** drop the battery as this can damage the internals
- **DO NOT** store the battery in a freezer or expose to extreme heat
- Battery should be fully charged before use

Battery Voltage

Battery voltage is displayed on the status bar at the top of every screen. A fully charged battery will indicate a voltage between **7.6 V and 8.1 V**. When the voltage drops below **7.0 V**, the battery icon will turn orange. If the voltage drops below **6.7 V**, a red low-battery icon will be displayed.



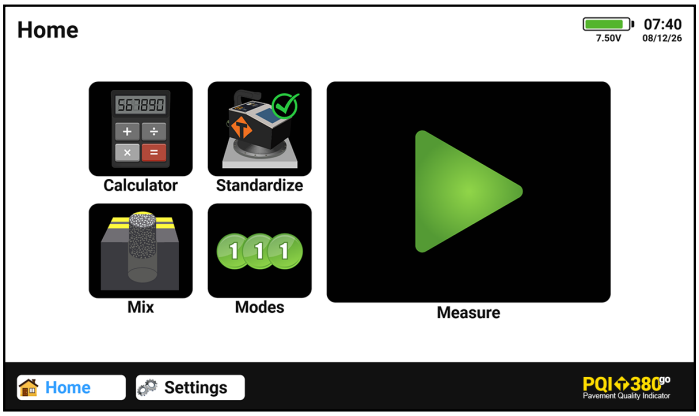
Low Battery Indicator



The gauge will continue to operate until the battery can not supply enough voltage to complete a measurement. Once this happens, the gauge will automatically shut down and will not restart until charged. **It is important to re-charge the battery after each use.**

Starting the Software





After powering on, the PQI 380go initializes for 30 seconds while displaying multiple splash screens. At the end of boot up process the Home screen is displayed. The Home screen will display five options, **Calculator, Standardize, Mix, Modes** and **Measure.**

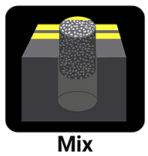
Home Menu



Calculator button opens the built in calculator screen



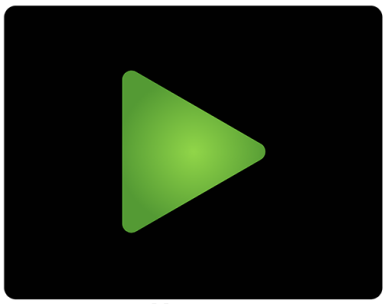
Standardize button opens the Standardization screen



Mix button opens the Mix screen

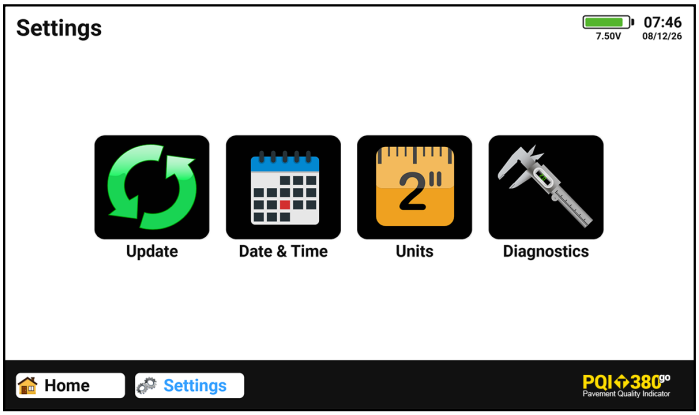


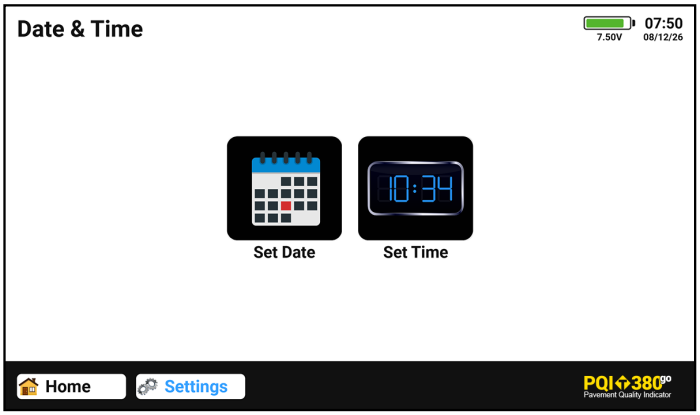
Modes button opens the Modes screen



Measurement button opens the measurement screen

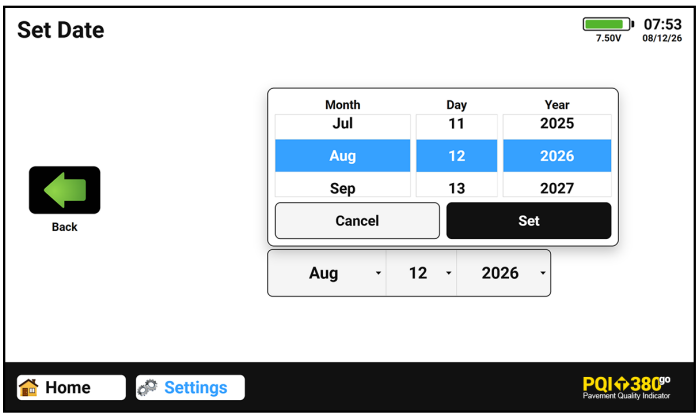
Local Time and Change Date Format

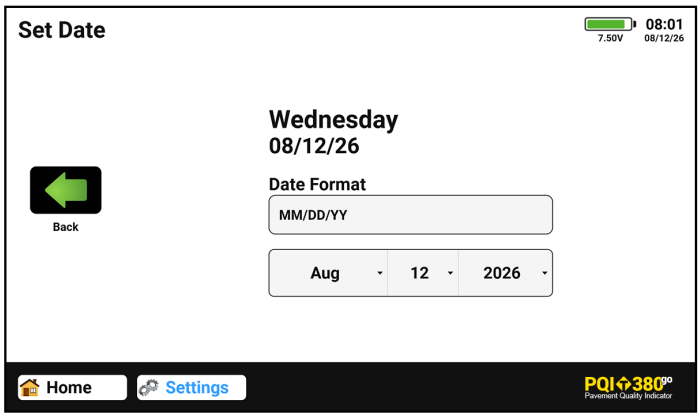




To change the date and time, tap the **Settings** button. From the Settings screen tap **Date & Time**. Once in the Date & Time screen tap either **Set Date** or **Set Time**.

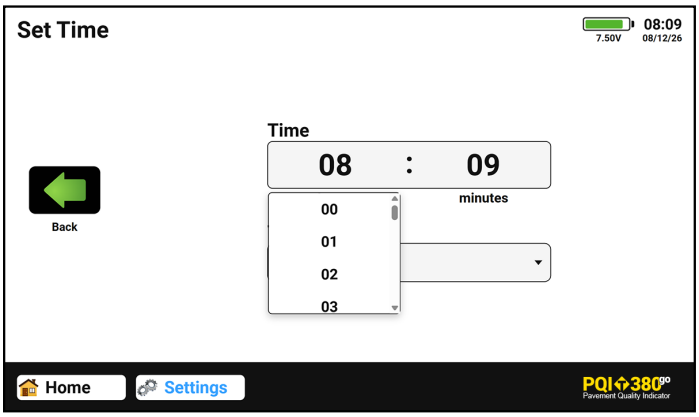
Set Date

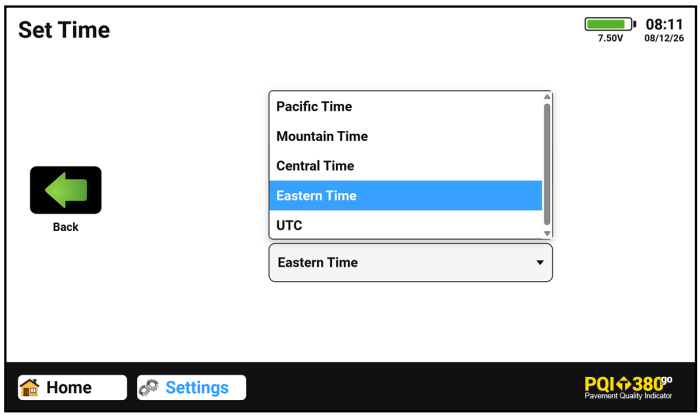




To set the date, tap the **Set Date** button. Then tap anywhere on the date bar and an input will pop up allowing you to set the month, day, and year. Once you are satisfied with your entry, tap the set button to accept. You can also toggle the date format from MM/DD/YY to DD/MM/YY by tapping the date format input. **(Note: the date will be preset at the factory)**

Set Time

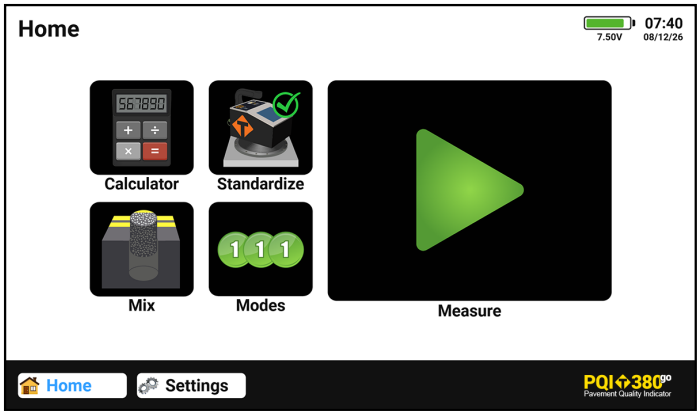


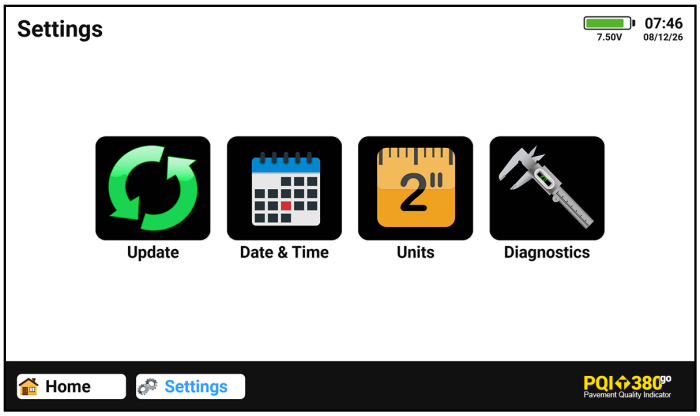


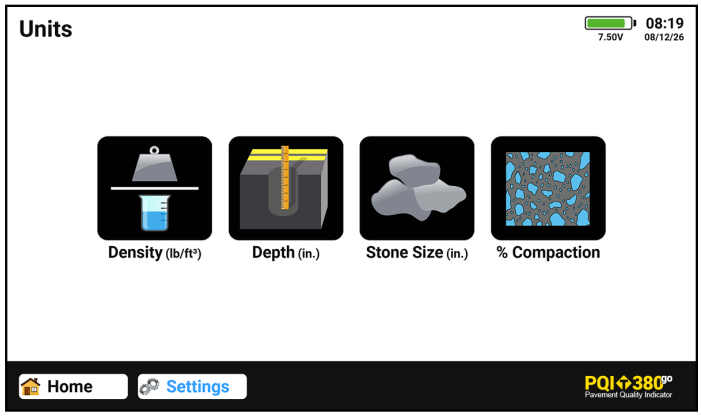
To set the time, tap the **Set Time** button. The time will display in 24 hour format along with the time zone. Tap anywhere on the time and a drop down menu will open allowing you to change the hours and minutes. If a change to the time zone is needed tap on the current time zone and a drop down menu will open. Tap the correct time zone and your selection will automatically be saved.

Select Measurement Units

From the Home screen, tap the **Settings** button on the task bar. Then tap **Units**.







Density, Depth, Stone Size and % Compaction can be toggled independently between system international (SI "metric") and U.S. customary units. For example, you can set the density to **lb/ft³** while depth is set to **inches** and stone size is set to **millimeters**. Simply tap any one of the icons to toggle the unit for that category.

Mix Details

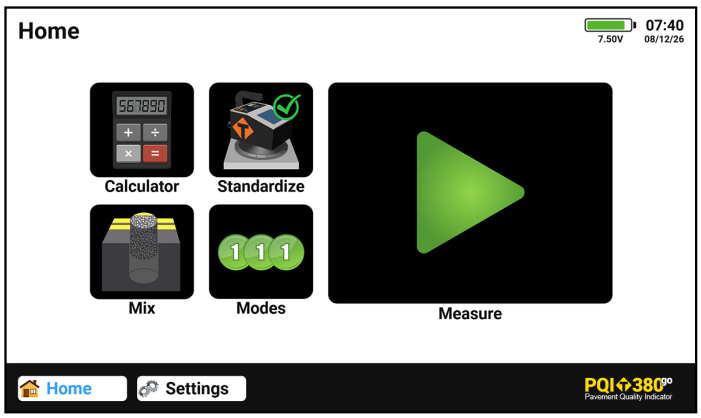
The density determined by the PQI 380go is highly material dependent so it is extremely important that the mix information for each mix design are input accurately into the gauge.

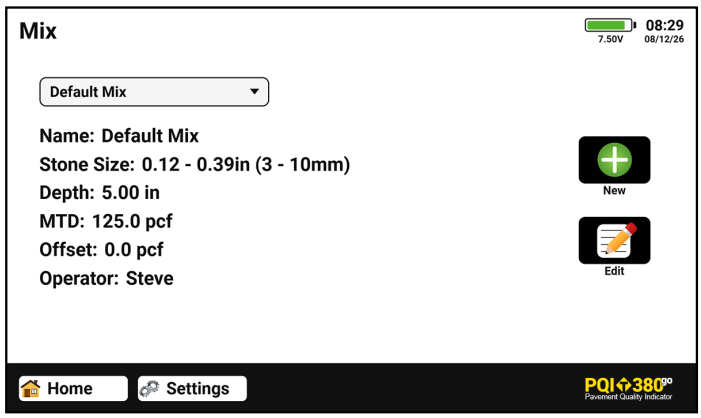
Note: Readings taken prior to accurately setting up the mix details section of the gauge will result in incorrect density and compaction results.

The PQI 380go is configured to store unique mixes that are identified by user entered descriptions. The default mix stored in a new PQI 380go will have generic mix name (i.e. Default Mix), the stone size will all be set to 9.5mm (0.375in.) at a depth of 5 inches with a Maximum Theoretical Density (MTD) of 125lb/ft³. **Readings taken prior to editing the default information on a mix that does not resemble the default information will result in incorrect density and compaction.**

Editing Mix Details

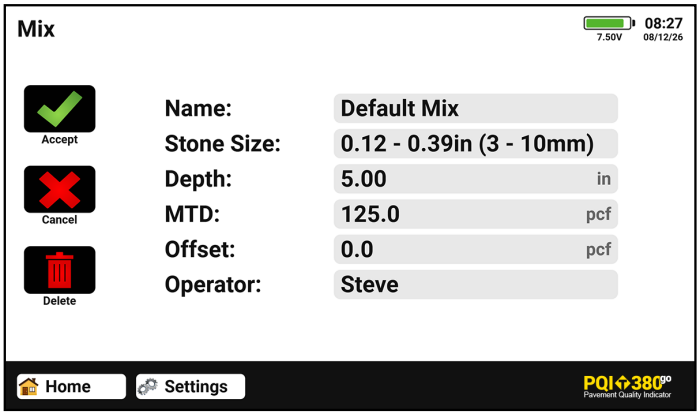
From the Home screen, tap **Mix** to enter the Mix screen. The active mix name is shown in the drop down menu. To edit the details of this mix, tap **Edit**.

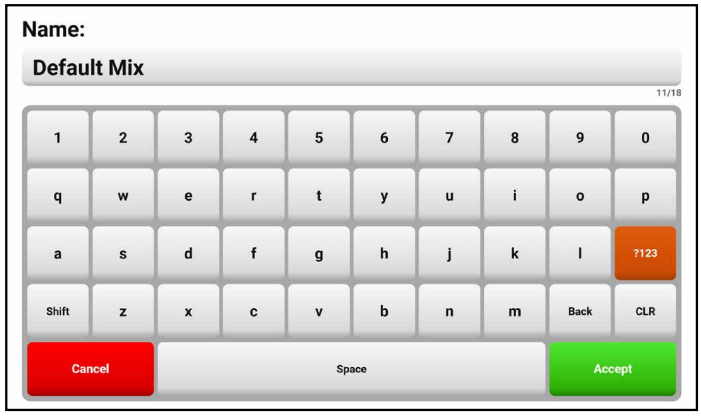




Editing Mix Details - Mix Name

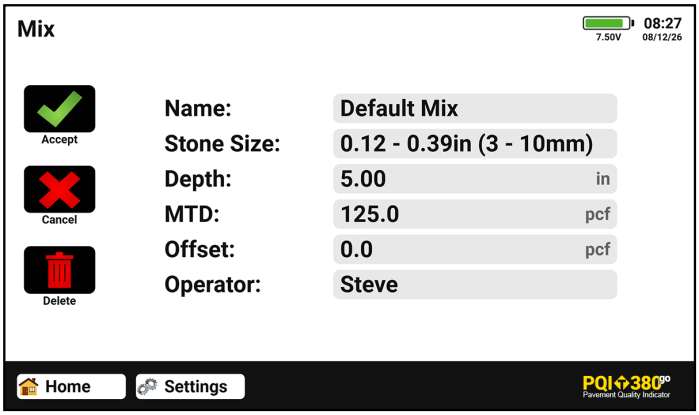
In the Edit Mix screen there are six fields labeled **Name, Stone Size, Depth, MTD, Offset** and **Operator**. By tapping one of these fields, you will enter the corresponding screen which will allow you to edit and save information for that mix value.

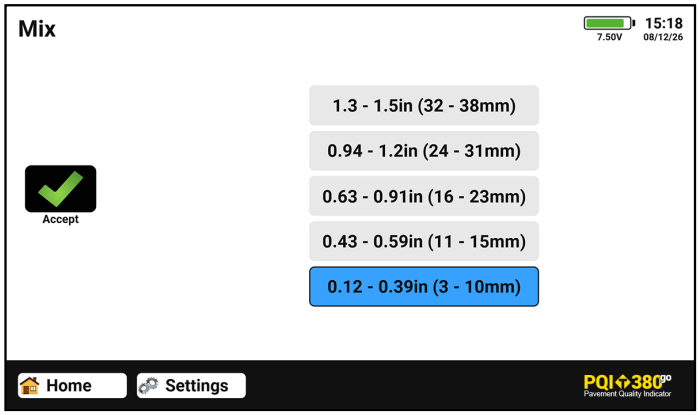




For example tap the gray shaded **Name** field, the keyboard will pop up allowing you to change the mix name. Tap **"CLR"** if you would like to start over with a new name. Tap **Shift** to toggle from lowercase to uppercase letters. Once editing has been completed, tap **Accept** to return to the Edit Mix screen.

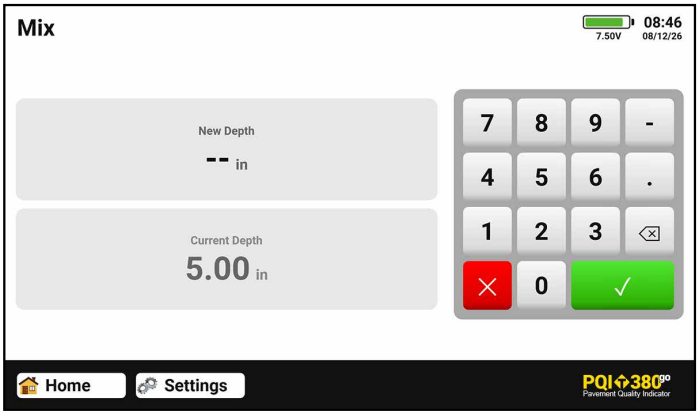
Editing Mix Details - Stone Size

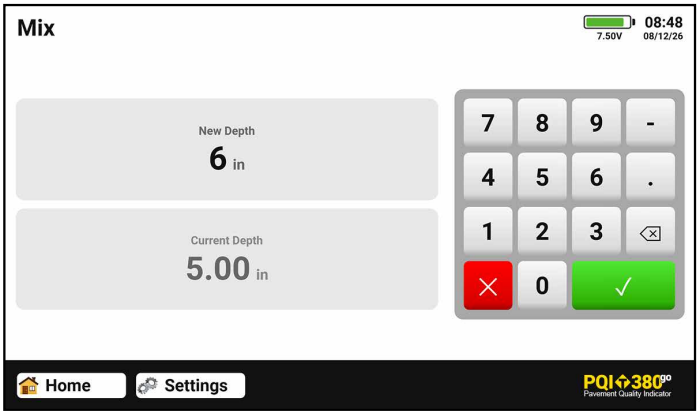




Tap the **Stone Size** field to enter the Stone Size screen. There are five options for aggregate sizes listed in inches (millimeters). If, however, you do not see your specific stone size listed choose the closest stone size available. For example, if your mix has an aggregate size of 0.79in (20mm), select **0.63 - 0.91in (16mm - 23mm)**. Tap **Accept** after you made your selection to return to the Edit Mix screen.

Editing Mix Details - Depth

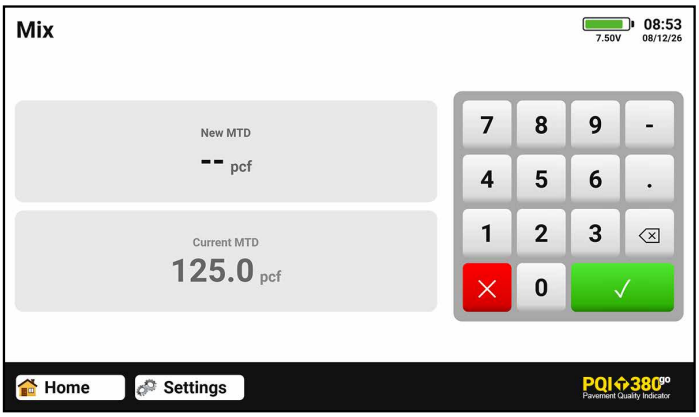


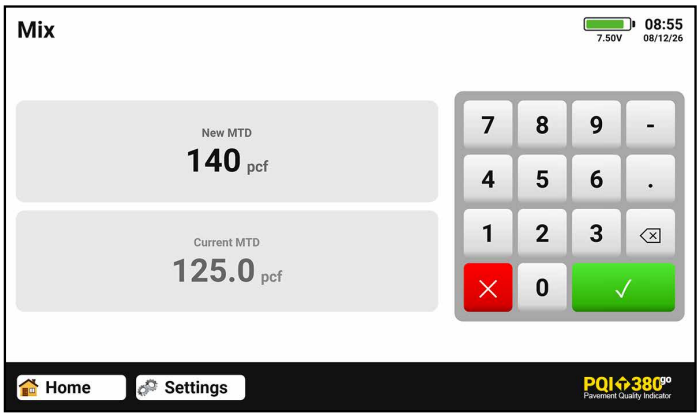


Tap the **Depth** field to enter the Edit Depth screen. Enter the depth of the asphalt mat on the numeric keypad in the units you previously selected for the depth. Once the correct value has been entered, tap the green **checkmark** to return to the Edit Mix screen. If you are satisfied with your new depth value, tap **Accept** to save your entry and return to the Mix screen.

An out of range Warning will display for depths entered outside of the following range:
0.75in - 6.0in (19.05mm - 152.4mm)

Editing Mix Details - Maximum Theoretical Density (MTD)

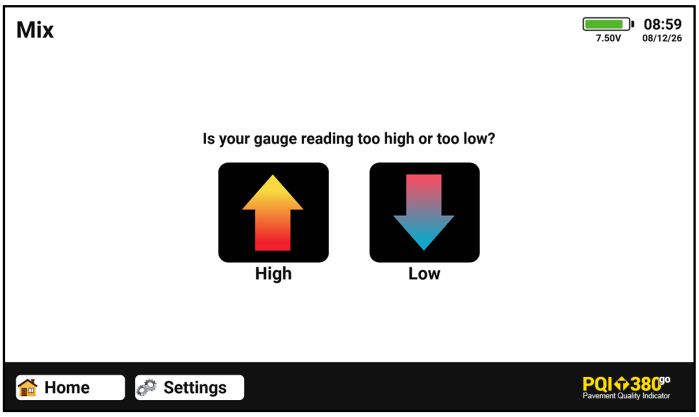


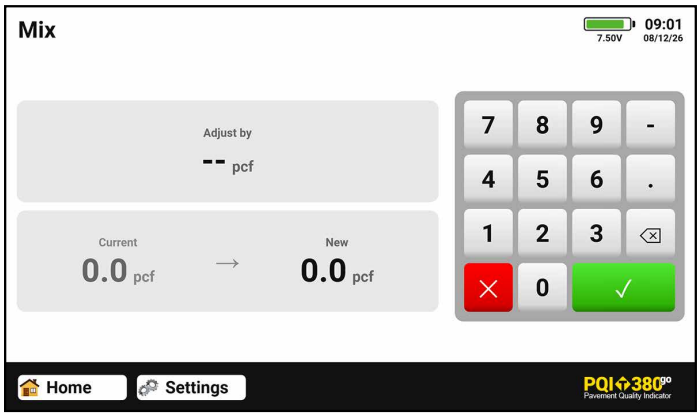


Tap the **MTD** field to enter the Edit MTD screen. Enter the Maximum Theoretical Density (MTD) of the mix on the numeric keypad in the units you previously selected. The MTD is provided from the asphalt mix designer and is a key value in determining the percent compaction. Once the correct value has been entered, tap the green **checkmark** to return to the Edit Mix screen. If you are satisfied with your new MTD value, tap **Accept** to save your entry and return to the Mix screen.

An out of range Warning will display for an MTD entered outside of the following range:
100lb/ft^3 - 200lb/ft^3 (1601.8kg/m^3 - 3203.7kg/m^3)

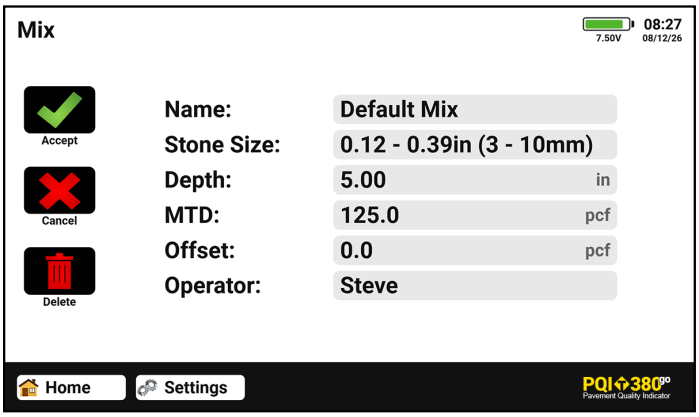
Editing Mix Details - Offset

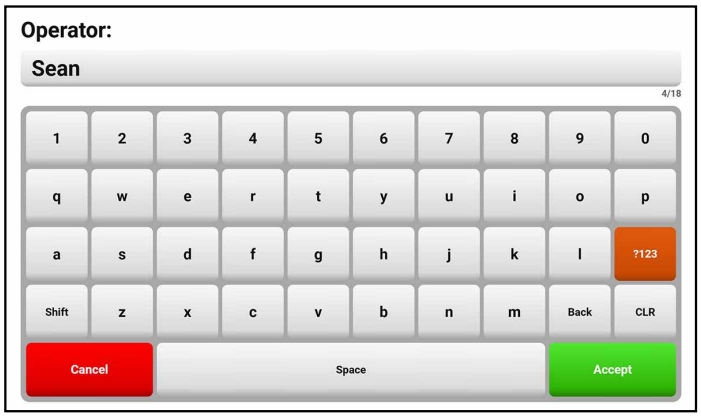




The offset for this material can be adjusted at a later time. **Measurements taken prior to adjusting the offset will not include the offset.** You will need to take a measurement to determine the offset. The offset feature will be revisited later in the calibration section of this document.

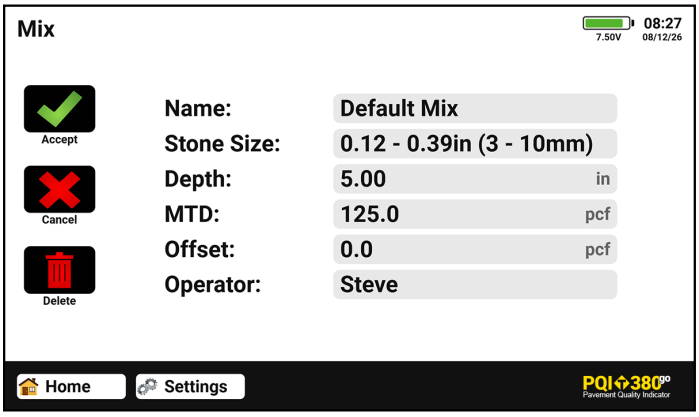
Editing Mix Details - Operator

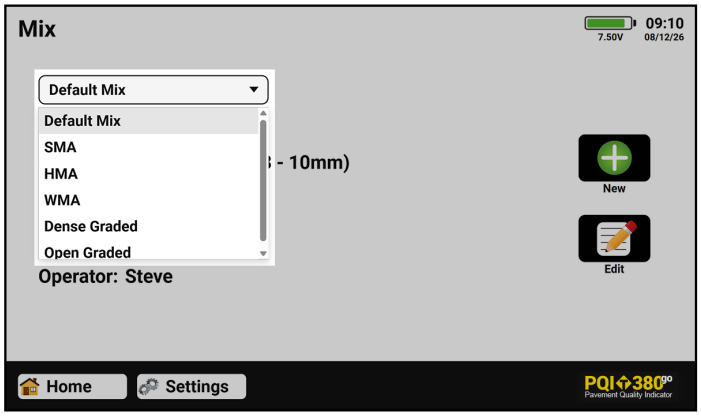




Tap the **Operator** field and a keyboard will pop up. Enter your name as the operator of the gauge, then tap **Accept**. The gauge will return to the Edit Mix screen. Verify all mix information on this screen. Once you have verified the results of the mix, tap **Accept** to save the mix.

Editing Mix Details





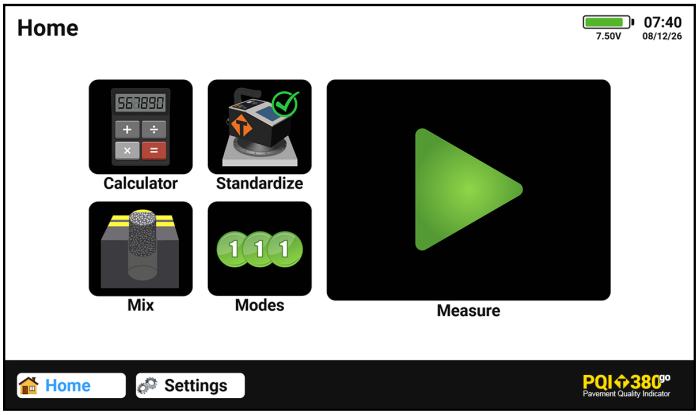
After tapping Accept from the Edit Mix screen the gauge will return to the **Mix** screen. Here you can tap the drop down menu to view and select different available mixes. **The CURRENT MIX that the gauge will use when taking readings is shown in the Name field. Prior to exiting this screen, be sure this is the mix you want and the information has been verified.** Tap **Home** on the task bar to return to the Home Screen. If the gauge is shutdown, the current mix prior to shutdown will remain the current mix when started back up.

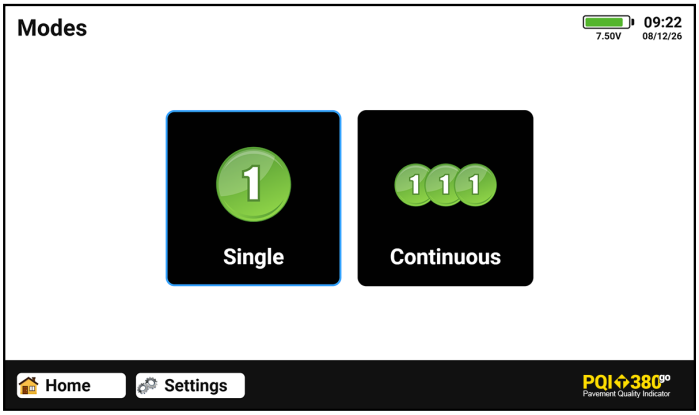
Reading Modes - Selection and Definition

The PQI 380go has two reading modes - **Single** and **Continuous**.

Data will NOT be saved on the PQI 380go, it is up to the operator to document the results.

Reading Mode Selection



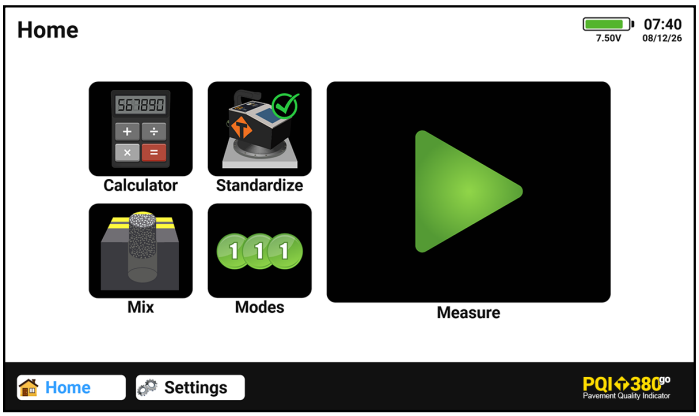


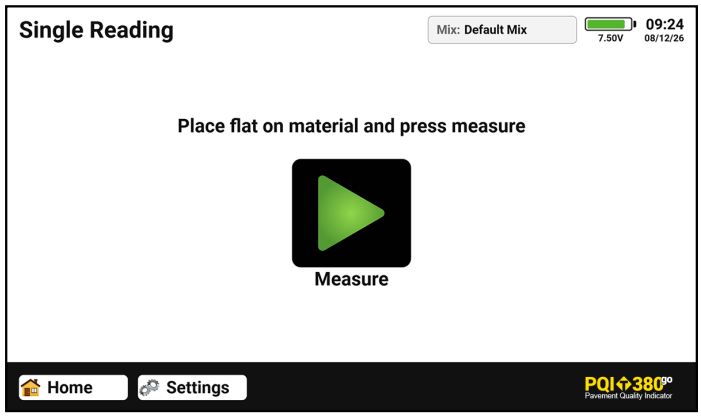
To select or change a reading mode tap **Modes** from the Home screen. Then tap the mode you would like to use. Once selected the active mode will be highlighted with a blue border and automatically redirect you to the measurement screen awaiting a reading. Continuous mode will start immediately.

Verify the mix name displayed in top of the measurement screen is the mix you intend to use. Do not touch or hover over the gauge when it is taking a reading. Place the gauge on a flat and dry surface whenever possible.



Reading Mode Definition

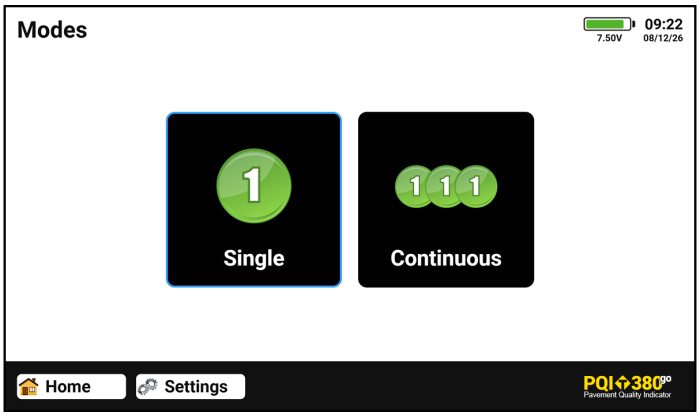


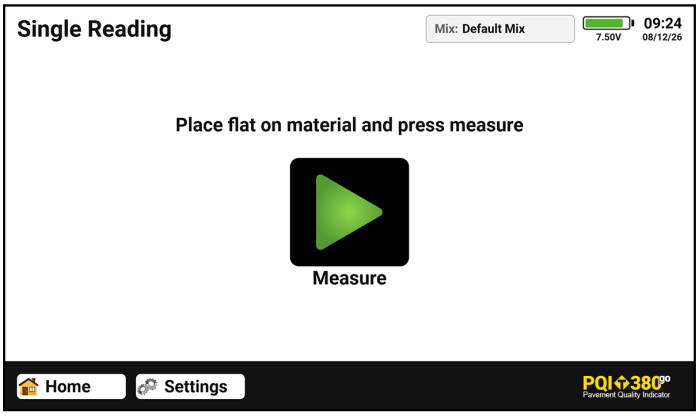


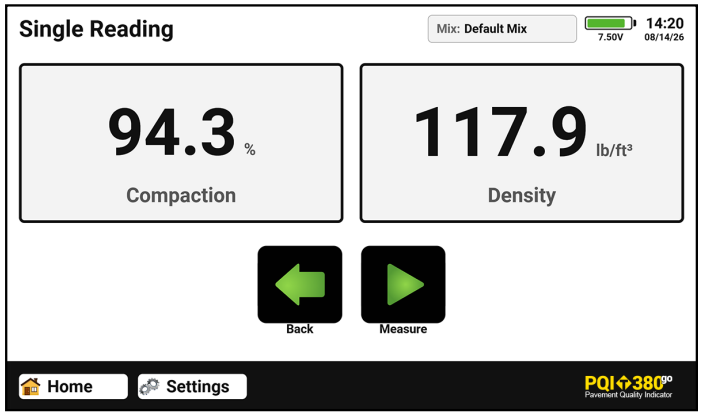
The last reading mode selected before the gauge is shut down will remain active. From the **Home** screen, tap **Measure** to enter the currently active reading mode to begin taking readings. The heading of the measurement screen will display the active mode **(Single Reading** or **Continuous Reading)**.

Single Reading Mode

From the Home screen, tap **Modes**, then tap **Single**. Press **Measure** on the Single Reading screen. The PQI 380go will display the results. **Data will NOT be saved on the PQI 380go, it is up to the operator to document the results.** Press **Measure** to take another reading or press **Back** to return to the Single Reading start screen.

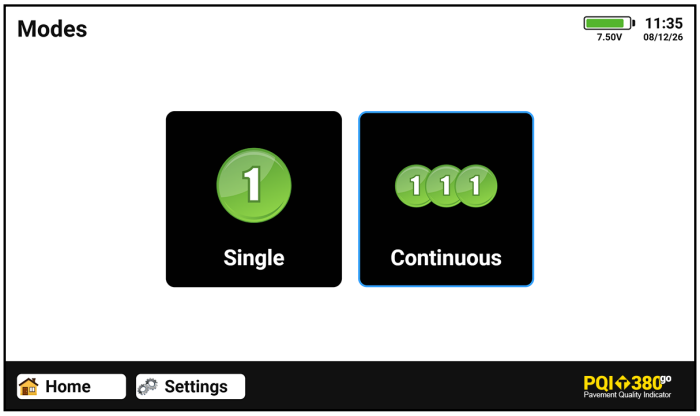


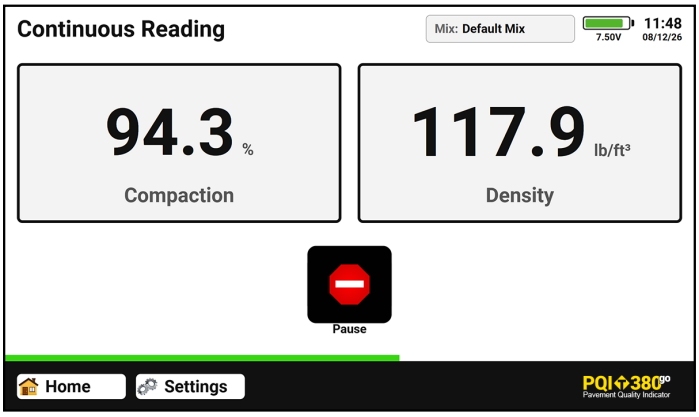


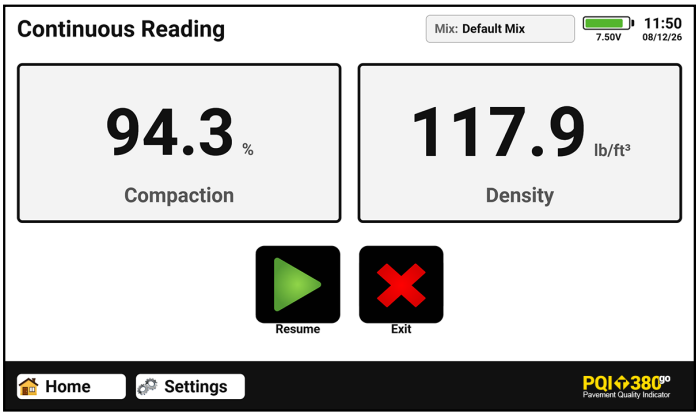


Continuous Reading Mode

From the Home screen, tap **Modes**, then tap **Continuous**. The PQI 380go will immediately start taking measurements every three seconds. **Data will NOT be saved on the PQI 380go, it is up to the operator to document the results.** Press **Pause** if needed to write down the results. Press **Resume** when you are ready to continue taking measurements or press **Exit** to return to the Home screen.







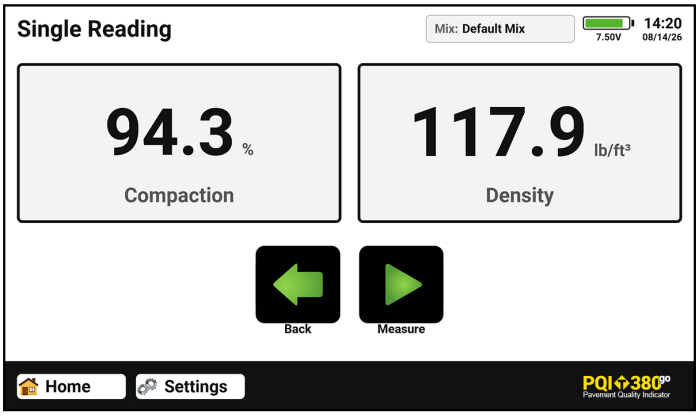
Calibration - Core Calibration Method

The calibration of the PQI 380go to each new mix is necessary for accurate and consistent readings. This is due to the various HMA mixtures being used in the field today. Changes in the aggregate type and size as well as changes in the binders produce a wide variety of electrical properties. Once calibrated to a standard (core) the PQI 380go will maintain optimal precision and accuracy.

For optimal results, the **core calibration method** is the most accurate method when calibrating the PQI 380go. The level of quality control from the plant to the mat will determine the number of test locations an operator should identify (one to five). The standard method for establishing a control strip as specified in **AASHTO TP 68** would be a total of five core locations. Remember the calibration will only be as good as the cores they are calibrated to. Therefore, the more test locations you have the more accurate the results.

Place the PQI 380go in the first location on the asphalt mat. Using the sensor plate as a guide, draw a circle around the gauge with a crayon marker. Using the **Single Mode** on the PQI 380go, tap **Measure** to take the first reading. **(DO NOT TOUCH OR HOVER OVER THE PQI WHILE IT IS TAKING A READING). Data will NOT be saved on the PQI 380go, it is up to the operator to document the results.**



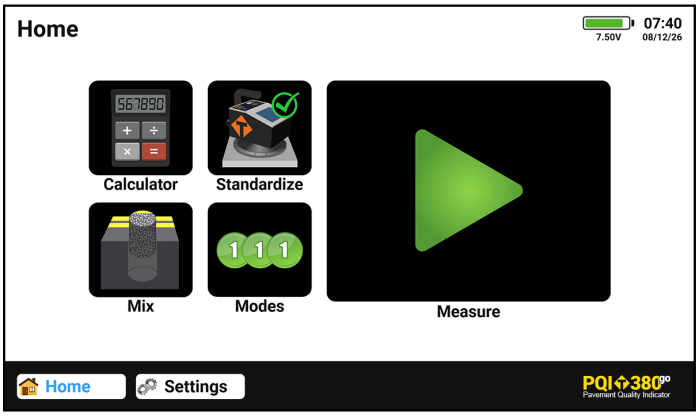


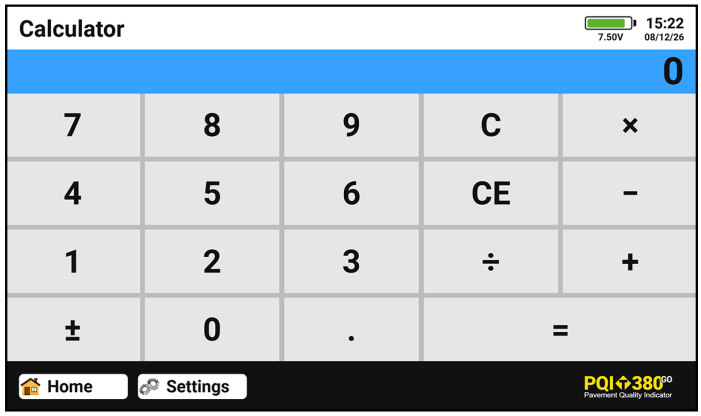
Move the PQI 380go approximately 2 inches up and to the right in the clover leaf formation. Tap **Measure** to begin the second reading. Continue through the clover leaf pattern until you have completed five individual readings.

Determine the average of these five individual readings. Write down the average PQI 380go result on the **Core Comparison Calibration Work Sheet** located at the back of this manual.

Calculating the Offset

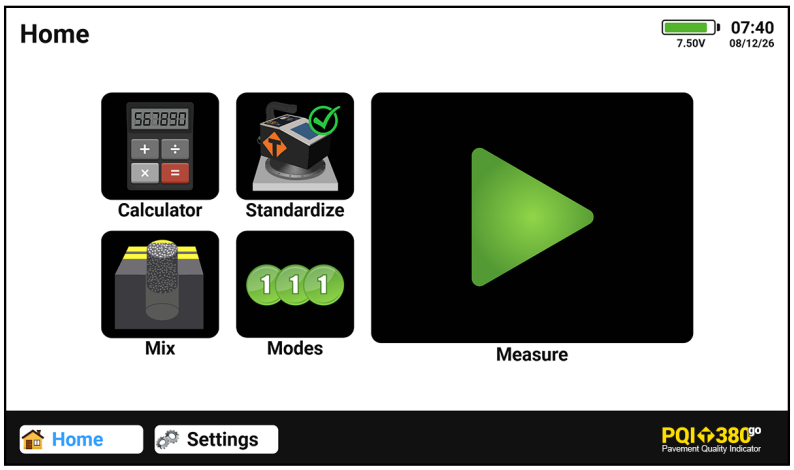
Direct percent compaction of the test mat is achieved by obtaining physical samples (cores) from the locations previously tested. Once the core density lab reports are received enter the value for each core on the **Core Comparison Work Sheet**. Calculate the difference between the average PQI 380go readings and the core density values. The five calculated differences (Core Density - PQI 380go Average Density) will then be averaged to obtain one difference. This difference will be the **Offset** for that specific mix that will be stored in the gauge for that mix. Therefore, you will be adjusting the value of the PQI 380go readings by that amount in order for the gauge to read the same as the core(s).





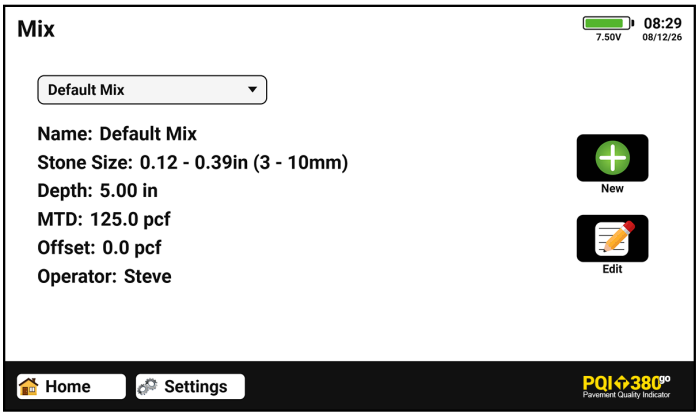
There is a built in calculator for your convenience if you do not have one handy. From the Home screen, tap **Calculator**. Once the numeric difference is calculated, determine whether the PQI 380go is reading **too high** or **too low**. For example, if the PQI reads **155lb/ft³**, and you would like it to read **150lb/ft³**, the PQI is reading too high and the **adjusted value for the offset is 5**.

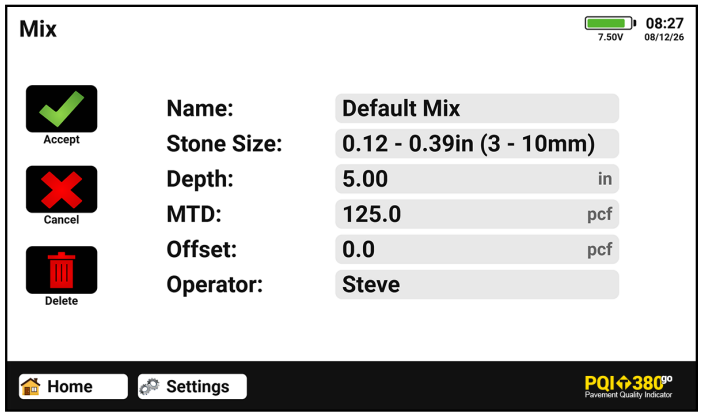
Editing Mix Details - PQI 380go Offset Setup



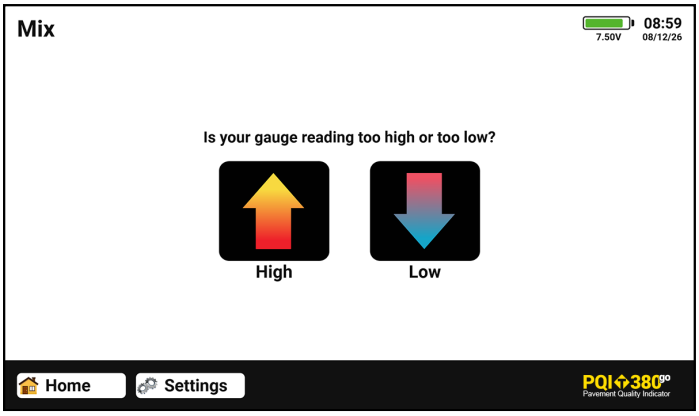
From the Home screen tap **Mix**.

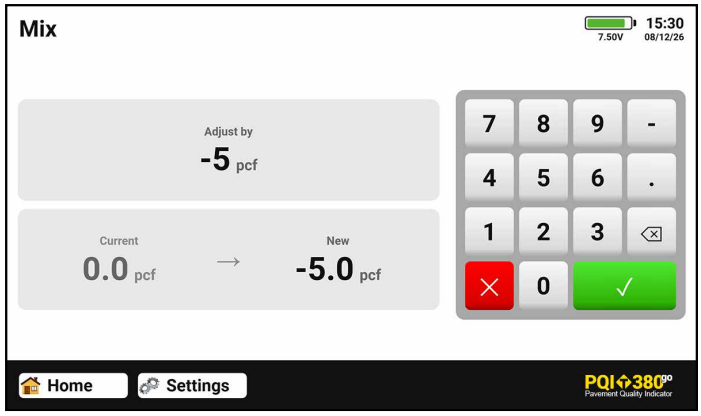
Editing Mix Details - PQI 380go Offset Setup



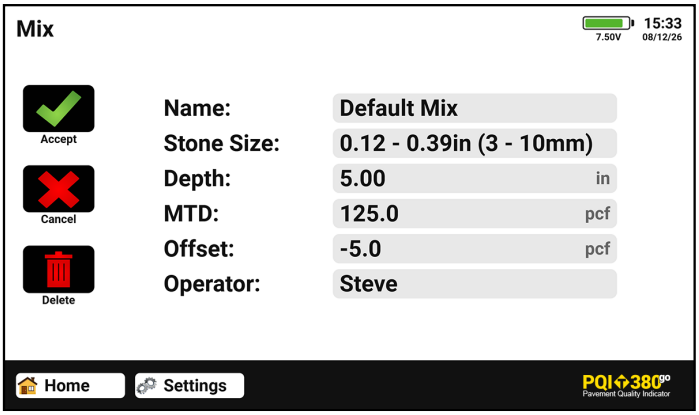


Be sure that the selected mix is the mix you wish to add the offset to. Tap **Edit**, then tap the gray shaded **Offset** field. In the previous step we determined that the PQI 380go is reading too high and the adjusted value for the offset is 5.





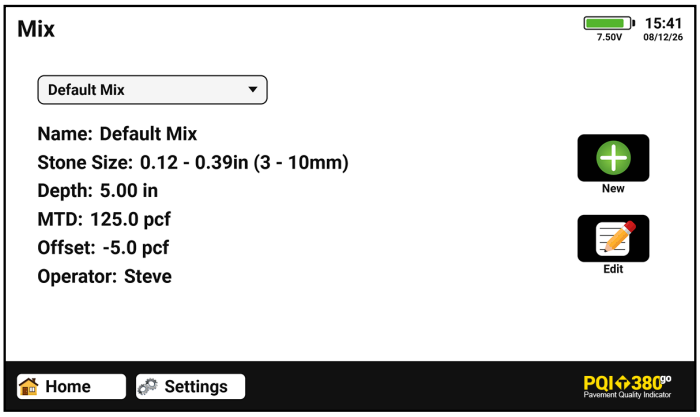
Tap **HIGH** to continue. The offset adjustment will display the current offset value. Enter the adjusted value for the offset which we previously determined would be 5, then tap the green **checkmark** to return to the **Edit Mix** screen.



In the Edit Mix screen, verify the updated offset information for that mix. Once satisfied, tap **Accept** again to save your results and return to the Mix screen.

Editing Mix Details - PQI 380go Offset Setup

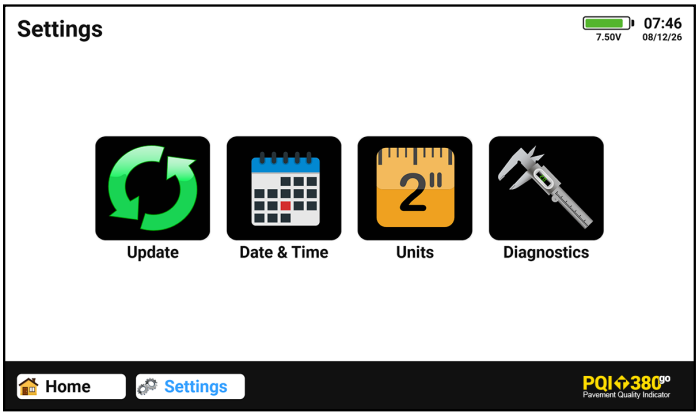
The adjusted value and new offset will be the same if the current offset was originally zero. If the current offset had been -1 and the adjusted value 5, then the new offset would be -6 for a PQI 380go that was reading too high.

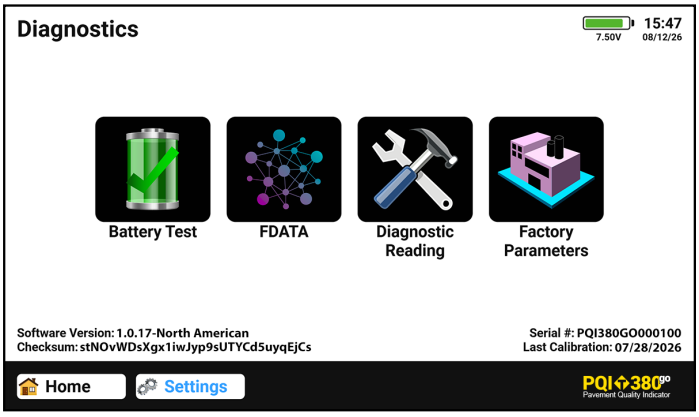


The offset for each mix can be adjusted at any time. Readings taken prior to a specific adjustment will not reflect that adjustment.

Maintenance and Troubleshooting

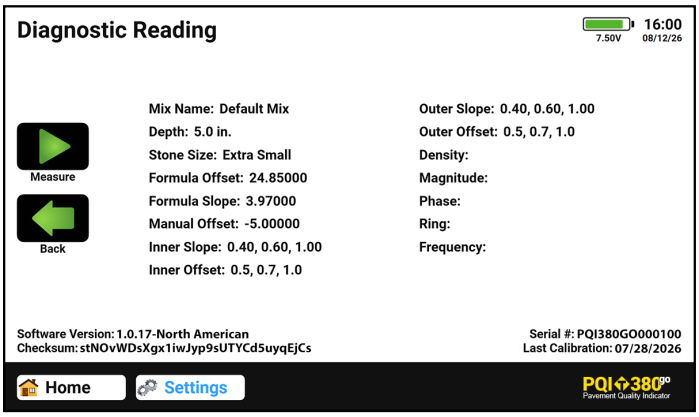
In the event that the PQI 380go were to malfunction an internal diagnostics mode can be activated to provide necessary technical information to the factory. If a repair is required, **we strongly recommended that authorized factory service be obtained**. Unauthorized repair or maintenance by the user during the warranty period will void the unit's warranty.

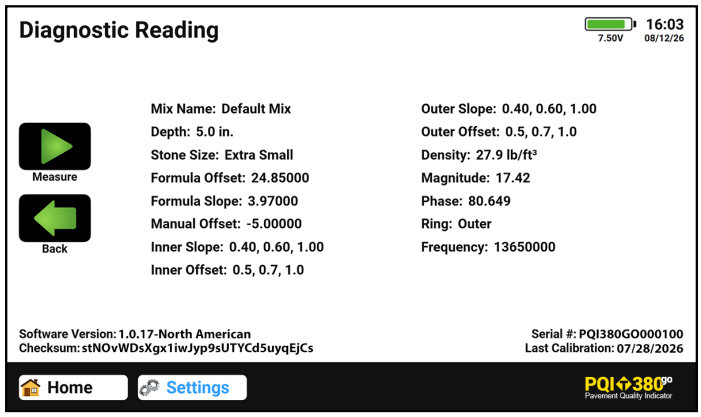




From the Settings screen, tap **Diagnostics**. From the Diagnostics menu, tap **Diagnostic Reading**.

Maintenance and Troubleshooting

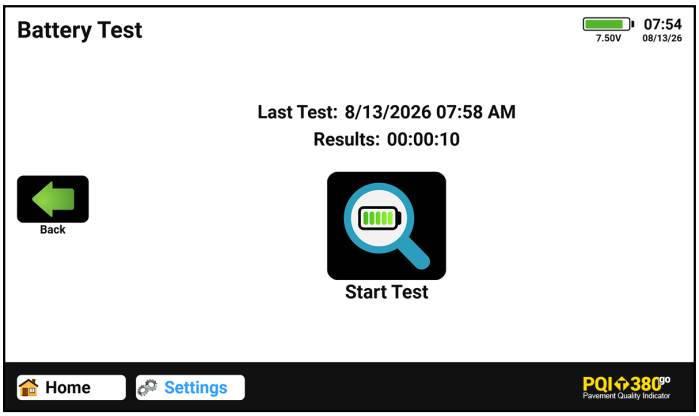


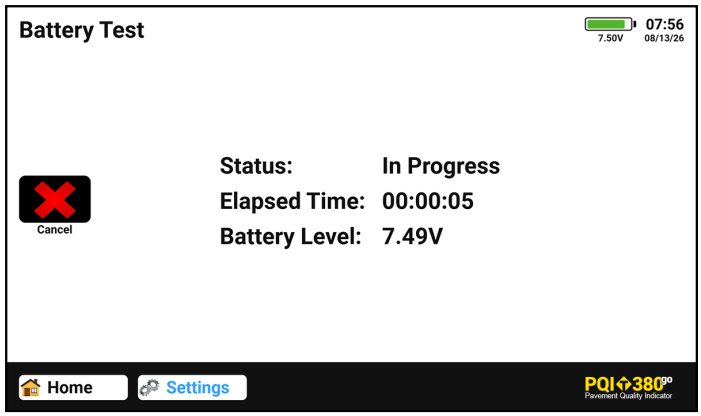


Place the gauge flat on the asphalt mat and tap **Measure**. Parameters from this display should be written down and relayed back to TransTech Systems or your distributor as a diagnostics tool.

Battery Test

The PQI 380go gauge allows you to test the battery pack for total usable charge duration with continuous use. A brand new battery pack should achieve a 7 hour 30 minute run time +/- 30 minutes.





From the Diagnostics screen, tap **Battery Test**. This will take you to the Battery Test screen where you can start the battery test by tapping **Start Test**. This screen will also show the results of the last test run time. When initiating the test ensure that the gauge has been fully charged. Allow the test to run until it automatically shuts down. Then charge the gauge fully and go back to the Battery Test screen to see the results.

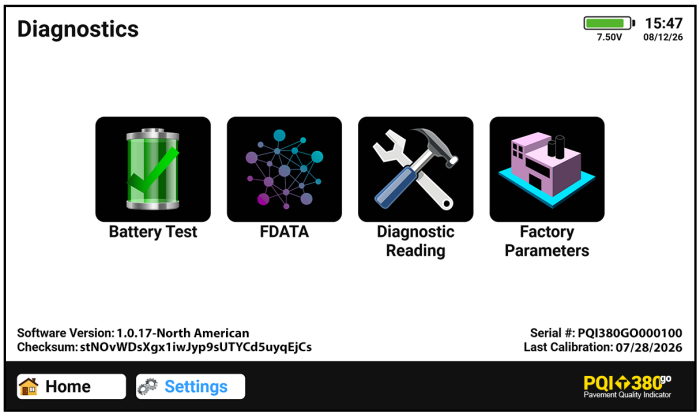
Factory Calibration

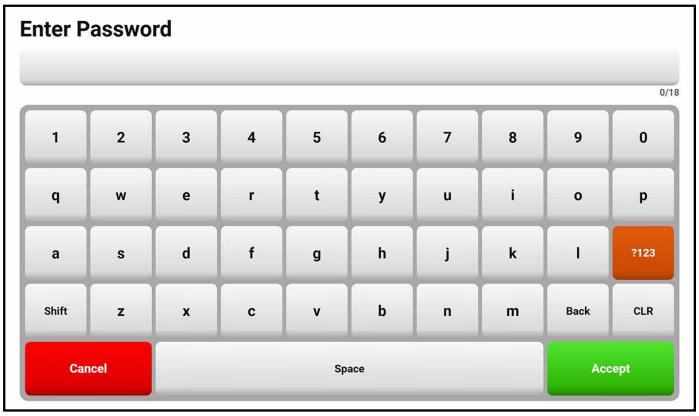
Yearly recalibration of the gauge is recommended to maintain consistency and accuracy for the electronic components.



If the gauge is past it's calibration due date a yellow warning screen will appear during start up to remind you to send the gauge in for calibration. Only TransTech Systems or authorized distributors are able to perform this type of calibration.

Factory Parameters





The **Factory Parameters** button is located in the diagnostics screen. This feature is used for TransTech Systems in house diagnostics. If this button is tapped the keyboard will pop up with a password prompt. In certain circumstances TransTech Systems may provide you with the password to assist them in diagnosing your gauge.

Maintenance

The PQI 380go is designed to require minimal maintenance and service. With proper care during use, along with annual preventive maintenance and calibration, the gauge should provide reliable, trouble-free operation for many years.

Keep the PQI 380 Plus bottom surface clean and dry:
For accurate readings, the PQI 380go should have a clean, dry, and smooth contact surface between the sensor plate and the mat. Wipe the sensor plate clean and dry after each reading. Before placing the PQI 380go on the mat, inspect the mat surface and remove any loose material that could prevent the gauge from seating properly. If asphalt begins to build up on the sensor plate, clean the surface using WD-40 or a citrus-based cleaner.

Touchscreen:
Do not clean the touchscreen with abrasive products.
Apply window cleaner to a soft rag and wipe the surface clean.

Factory Calibration:
Annual recalibration of the gauge is recommended to maintain the consistency and accuracy of its electronic components.

Battery Care Tips:
- Whenever uncertain about the battery charge level or condition, recharge it.
- The battery will self-discharge and therefore should NOT be left uncharged for more than 30 days.
- An occasional complete discharge followed by a full recharge is recommended.
- NEVER drop the battery as this can damage the internals.
- DO NOT store the battery in a freezer or expose to extreme heat
- Battery should be fully charged before use

Troubleshooting

The chart below provides guidance to a few suspect conditions.

Problem	Remedy
Buttons Sticking / Frozen Software	Repress the frozen button firmly. Be sure not to drag your finger from one button to the next. If this does not work, reset the gauge by turning it off and then back on.
Incorrect Density Reading	**Clean off sensor plate** **Check Current Mix Details-** be sure you have selected the proper aggregate size and entered all details correctly **Check Calibration-** check your calculations for the adjusted value of the offset and that it was entered correctly **Call Factory**
Battery Problems / Gauge Keeps Shutting Down / Gauge Will Not Come On	**Ensure battery pack is plugged in** **Check battery voltage-** plug gauge into charger and wait for the green LED to illuminate to indicate a full charge **Call Factory**

Standardization

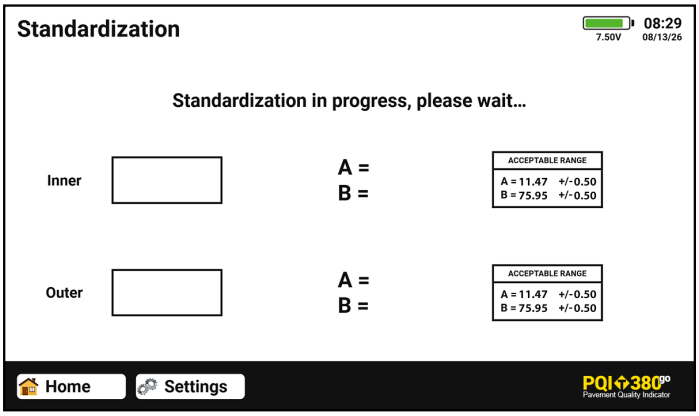
To verify that the PQI 380go continues to provide consistent measurements, perform a daily measurement on a reference material and monitor the results for any unacceptable variation over time. A metal reference plate is installed in the bottom of the PQI 380go carrying case and can be used for this purpose.

Although this verification is referred to as gauge standardization, the standardization results do not affect the gauge's measurements. Instead, they serve only to alert the user to changes in the gauge's operating performance. Any unexpected changes in the standardization values should be documented and discussed with the TransTech Systems service team.

Standardizations are performed **inside the carry case**. Located at the bottom of the case is a steel plate. **Position the gauge on top of the plate inside the case**. **Be sure the gauge screen is facing you, if it is placed in the case backwards the handle will prevent proper seating you will get a false reading.** The case should not be on top of or around any large metal objects. From the Home screen, tap the **Standardize** button. Tap **Start**, to begin the standardization.

Do not touch the gauge while it is standardizing.

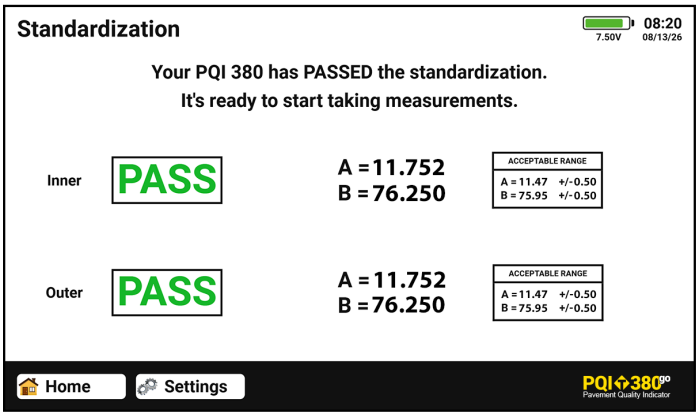


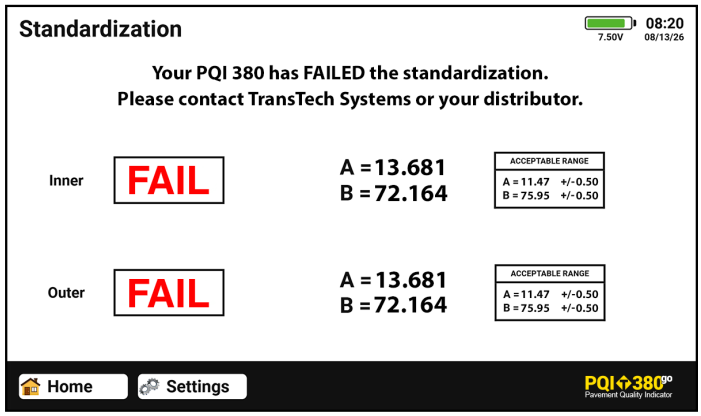


The acceptable range for the A and B values are displayed on the screen, however, the PQI 380go will respond with a **PASS** or **FAIL** based on these ranges automatically.

Standardization

The result should read **PASS** for both inner and outer. If the result indicates a **FAIL** for either inner or outer, call TransTech Systems or your distributor.





If you receive a FAIL, prior to calling TransTech Systems try the following:

1. Reposition the gauge / case to another location (outside preferably)
2. Be sure that you are testing the gauge inside the carry case that was provided with the PQI. The gauge and case are paired during production to determine the standardization.
3. Do not place the case on the back or near an automobile as well as any other large metal objects and / or wires
4. Be sure that no foreign objects have adhered to the bottom of the gauge or on the steel plate
5. Shut off all cell phones within 10ft of the gauge

WARNING: THE STANDARDIZATION OF THE PQI 380go WILL ONLY WORK WITH THE PLATE IT HAS BEEN ISSUED TO. IF YOU RECEIVE A FAIL RESULT, CHECK TO BE SURE THE SERIAL NUMBERS LOCATED ON THE GAUGE AND THE STANDARDIZATION PLATE ARE THE SAME.

WARNING: THE STANDARDIZATION OF THE PQI 380go MAY GIVE A FALSE “FAIL” IF TEST IS TAKEN TOO CLOSE TO METAL OBJECTS, POWER LINES OR OTHER ELECTRICAL DEVICES THAT MAY BE TOO CLOSE TO THE GAUGE. TURN THE GAUGE OFF AND MOVE THE GAUGE AT LEAST 10 FEET FROM THESE OBJECTS. IF AFTER THESE PRECAUTIONS ARE TAKEN THE GAUGE STILL DISPLAYS A “FAIL” CONTACT YOUR DISTRIBUTOR.

Measurement Table

Company Name _____ Date _____

Job Site _____

Asphalt Mix _____ MTD _____

Core Comparison Calibration Work Sheet

1. Enter the core density readings and the average PQI 380go readings for each location.
2. Subtract the PQI readings from the core density for each location.
3. Add the differences for each location = total difference.
4. Calculate the average difference (total difference ÷ 5). This is the adjusted value for the offset.

Location	1	2	3	4	5
Core Density					
PQI Readings					
Difference					

Total Difference	
Average Difference (Adjusted Value Offset)	

Measurement Table

Company Name _____ Date _____

Job Site _____

Asphalt Mix _____ MTD _____

Core Comparison Calibration Work Sheet

1. Enter the core density readings and the average PQI 380go readings for each location.
2. Subtract the PQI readings from the core density for each location.
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Core Density					
PQI Readings					
Difference					

Total Difference	
Average Difference (Adjusted Value Offset)	

Measurement Table

Company Name _____ Date _____

Job Site _____

Asphalt Mix _____ MTD _____

Core Comparison Calibration Work Sheet

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Location	1	2	3	4	5
Core Density					
PQI Readings					
Difference					

Total Difference	
Average Difference (Adjusted Value Offset)	

Measurement Table

Company Name _____ Date _____

Job Site _____

Asphalt Mix _____ MTD _____

Core Comparison Calibration Work Sheet

1. Enter the core density readings and the average PQI 380go readings for each location.
2. Subtract the PQI readings from the core density for each location.
3. Add the differences for each location = total difference.
4. Calculate the average difference (total difference ÷ 5). This is the adjusted value for the offset.

Location	1	2	3	4	5
Core Density					
PQI Readings					
Difference					

Total Difference	
Average Difference (Adjusted Value Offset)	

TransTech Systems Product Warranty

The Company warrants to the Purchaser that the product delivered hereunder will be free from defects in material or workmanship and be the kind and quality designated or specified in the contract or purchase order. This warranty shall apply only to defects appearing within one (1) year from the date of shipment by the Company.

If the product delivered hereunder does not meet the above warranty and if the Purchaser promptly notifies the Company, the Company shall thereupon correct any defect, including nonconformance with the specifications, either (at the Company's option) by repairing any defective or damaged parts of the product, replacing the product, or by making available the necessary repaired or replacement parts.

The liability of the Company under this warranty, for any loss, whether the claim is based on contract or negligence, shall not in any case exceed the cost of correcting defects in the product as herein provided, and upon the expiration of the warranty period, all such liability shall terminate. The foregoing shall constitute the exclusive remedy of the Purchaser and the exclusive liability of the Company. The foregoing warranty is exclusive and in lieu of all other warranties, whether written, oral, implied or statutory.

No warranty of merchantability or of fitness for purpose shall apply. Unauthorized service shall void this warranty.

TransTech Systems Product Non-Warranty Return Policy (Domestic Sales Only)

Non-warranty returns for TransTech Systems Inc. products must be made within ten (10) days from the original date of shipment, unless otherwise indicated. Returned products must be in the original packaging, unused and in undamaged condition. Proof of purchase is required. Upon receipt of the product TransTech will inspect the product to the above mentioned criteria. TransTech Systems reserves the right to apply a restocking fee.

Unused products will be issued a credit to the Purchaser's account that was used to purchase the product. TransTech will not credit prepaid shipping cost. The original packing slip or invoice is required to be sent back with the product to be returned.

The Purchaser is responsible for shipping the product back to TransTech, carefully package the item(s) and include the packing slip and return manufacturing authorization number on the package. Prepay shipping is required – TransTech will not accept C.O.D.s.

Returns will be credited within 10 working days.

Proper Process of Warranty or Non-Warranty Shipments to TransTech Systems

For product returns (warranty or non-warranty), please follow the instructions below to assure prompt handling:

Call us for authorization (518-370-5558 or 800-724-6306), obtain a return manufacturing authorization number (RMA), and the return shipping address.

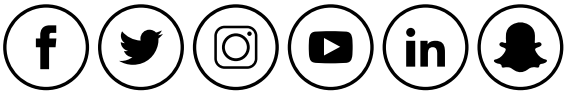
Indicate to our representative the reason for returning the product.

For Warranty Returns, Purchaser is responsible for shipping to TransTech's office.
TransTech will pay ground shipping return to the Purchaser.

TransTech Systems, Inc.

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Toll Free: 1-800-724-6306
Fax: 518-370-5538
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